

## The geographic distribution of unhealthy living characteristics according to the American Nations model: Cultural factors warranting attention

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### ARTICLE INFO

#### Keywords:

Healthy living  
Physical inactivity  
Obesity  
Diabetes  
Chronic disease

The United States (U.S.) is a prime example of a country experiencing parallel pandemics in unhealthy living characteristics (e.g., sedentary lifestyle, poor nutrition, and excess body mass) and chronic disease.<sup>1,2</sup> These twin pandemics are linked, as unhealthy living characteristics are primary risk factors for a range of chronic conditions, including diabetes mellitus (DM), cardiovascular disease (CVD), and certain forms of cancer.<sup>3</sup> The combined effects can have tragic consequences, as we learned during the coronavirus disease 2019 (COVID-19) pandemic: the risk of hospitalizations and death due to viral infection were significantly greater among those who presented with unhealthy living characteristics and/or were diagnosed with one or more chronic health condition.<sup>2,4,5</sup>

In previous publications, we have drawn attention to the heterogeneous geographic distribution of unhealthy living characteristics and chronic disease in the U.S., clearly demonstrating the presence of unhealthy living-chronic disease *hot spots* and the fact that these *hot spots* coexist is not surprising.<sup>2</sup> We also showed that COVID-19 mortality *hot spots* align with areas that have the highest prevalence of unhealthy lifestyle characteristics and chronic disease.<sup>2</sup> There is also a clear social justice aspect, as underrepresented individuals in underserved communities demonstrate higher levels of unhealthy living characteristics and chronic disease as well as poorer COVID-19 outcomes.<sup>6</sup>

The U.S. is experiencing a sustained downward spiral in population

health (e.g., chronic disease and poor outcomes with viral pandemics) primarily because it has high levels of unhealthy living characteristics. An obvious remedy would be for relevant stakeholders to promote healthy living (HL) behaviors from the population and individual level, and for the healthcare system to better practice HL medicine (HLM).<sup>1,7</sup> Specifically, all individuals should *move more and sit less* every day,<sup>8</sup> consume a healthful diet, not smoke, and strive toward minimizing excess body mass. The importance of HL is not a new concept. We have a well-established scientific understanding of the devastating effects of a sedentary lifestyle, consumption of a poor diet and obesity; conversely, the positive impact of HL is abundantly clear.<sup>9,10</sup> Despite this, our society has failed to reverse course and has yet to substantially improve HL characteristics.<sup>11,12</sup>

One of the reasons the U.S. has not gained momentum in improving lifestyle behaviors is the approach commonly used to promote HL (from a public health perspective) and how we prescribe HLM (from a healthcare perspective). There is certainly support for individualizing the HL message,<sup>13</sup> but in practice we continue to follow a *one-size-fits-all* approach that holds that exercising, eating nutritious foods, and maintaining a healthy body weight are good for you no matter who you are.<sup>14</sup> But another approach would be more effective because it would optimize the likelihood for adoption and adherence: tailoring the message and approach according to the unique characteristics of a given

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<https://doi.org/10.1016/j.pcad.2023.07.002>

Available online 5 July 2023

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**Table 1**

The identities of the ‘American Nations’.

|                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yankeedom</b> (pop. 52.6 million)                                                                                                                                                                                                                                                                                                                                                                             |
| Founded by Puritans who sought to perfect earthly society through social engineering, individual denial for common good, and the assimilation of outsiders. The common good – ensured by popular government – took precedence over individual liberty when the two were in conflict.                                                                                                                             |
| <b>New Netherland</b> (pop. 20.9 million)                                                                                                                                                                                                                                                                                                                                                                        |
| Dutch-founded and retains characteristics of 17th century Amsterdam: a global commercial trading culture, materialistic, multicultural, and committed to tolerance and the freedom of inquiry and conscience.                                                                                                                                                                                                    |
| <b>Tidewater</b> (pop. 13.2 million)                                                                                                                                                                                                                                                                                                                                                                             |
| Founded by lesser sons of landed gentry seeking to recreate the semi-feudal manorial society of English countryside. Conservative with strong respect for authority and tradition, this culture is rapidly eroding because of its small physical size and the massive federal presence around D.C. and Hampton Roads.                                                                                            |
| <b>Greater Appalachia</b> (pop. 61.5 million)                                                                                                                                                                                                                                                                                                                                                                    |
| Settlers overwhelmingly from war-ravaged Northern Ireland, Northern England and Scottish Lowlands were deeply committed to personal sovereignty and intensely suspicious of external authority.                                                                                                                                                                                                                  |
| <b>The Midlands</b> (pop. 35.9 million)                                                                                                                                                                                                                                                                                                                                                                          |
| Founded by English Quakers, who believed in humans’ inherent goodness and welcomed people of many nations and creeds. Pluralistic and organized around the middle class; ethnic and ideological purity never a priority; government seen as an unwelcome intrusion.                                                                                                                                              |
| <b>Deep South</b> (pop. 48.0 million)                                                                                                                                                                                                                                                                                                                                                                            |
| Established by English Barbadian slave lords who championed classical republicanism modeled on slave states of the ancient world, where democracy was the privilege of the few and subjugation and enslavement the natural lot of the many.                                                                                                                                                                      |
| <b>New France</b> (pop. 2.6 million)                                                                                                                                                                                                                                                                                                                                                                             |
| An enclave of a larger culture encompassing Quebec and parts of Atlantic Canada, the legacy culture was consensus driven, tolerant, and comfortable with government involvement in the economy, though these characteristics appear to have collapsed in much of Cajun country in recent decades.                                                                                                                |
| <b>El Norte</b> (pop. 34.3 million)                                                                                                                                                                                                                                                                                                                                                                              |
| Borderlands of Spanish-American empire, so far from Mexico City and Madrid that it developed its own characteristics: independent, self-sufficient, adaptable, and work-centered. Often sought to break away from Mexico to become an independent buffer state, annexed into U.S. instead.                                                                                                                       |
| <b>Left Coast</b> (pop. 18.8 million)                                                                                                                                                                                                                                                                                                                                                                            |
| Founded by New Englanders (who came by ship) and farmers, prospectors and fur traders from the lower Midwest (by wagon), it’s a fecund hybrid of Yankee utopianism and the Appalachian emphasis on self-expression and exploration.                                                                                                                                                                              |
| <b>Far West</b> (pop. 30.3 million)                                                                                                                                                                                                                                                                                                                                                                              |
| Extreme environment stopped eastern cultures in their path, so settlement largely controlled by distant corporations or federal government via deployment of railroads, dams, irrigation, and mines; exploited as an internal colony, with lasting resentments.                                                                                                                                                  |
| <b>First Nation</b> (pop. 61 thousand)                                                                                                                                                                                                                                                                                                                                                                           |
| Native American groups that generally never gave up their land by treaty and have largely retained cultural practices and knowledge that allow them to survive in this hostile region on their own terms. The nation is now reclaiming its sovereignty, having won considerable autonomy in Alaska and Nunavut and a self-governing nation state in Greenland that stands on the threshold of full independence. |

audience.<sup>15,16</sup> An important aspect to consider in this regard are the unique cultural identities of the U.S.’s distinct regions. The current commentary provides a visual depiction of the differences in a primary unhealthy living characteristic, opportunities to improve behaviors, and care of diseases, such as obesity and DM according to the *American Nations* model, developed in by Colin Woodard in 2011<sup>17</sup> and applied to various historical and public policy issues by the Nationhood Lab, Pell Center for International Relations and Public Policy, Salve Regina University.<sup>18</sup>

Cultural geographers have long recognized First Settler effects on the characteristics of national cultures, with Wilbur Zelinsky’s<sup>19</sup> Doctrine of First Effective Settlement arguing that “the dominant culture of a given

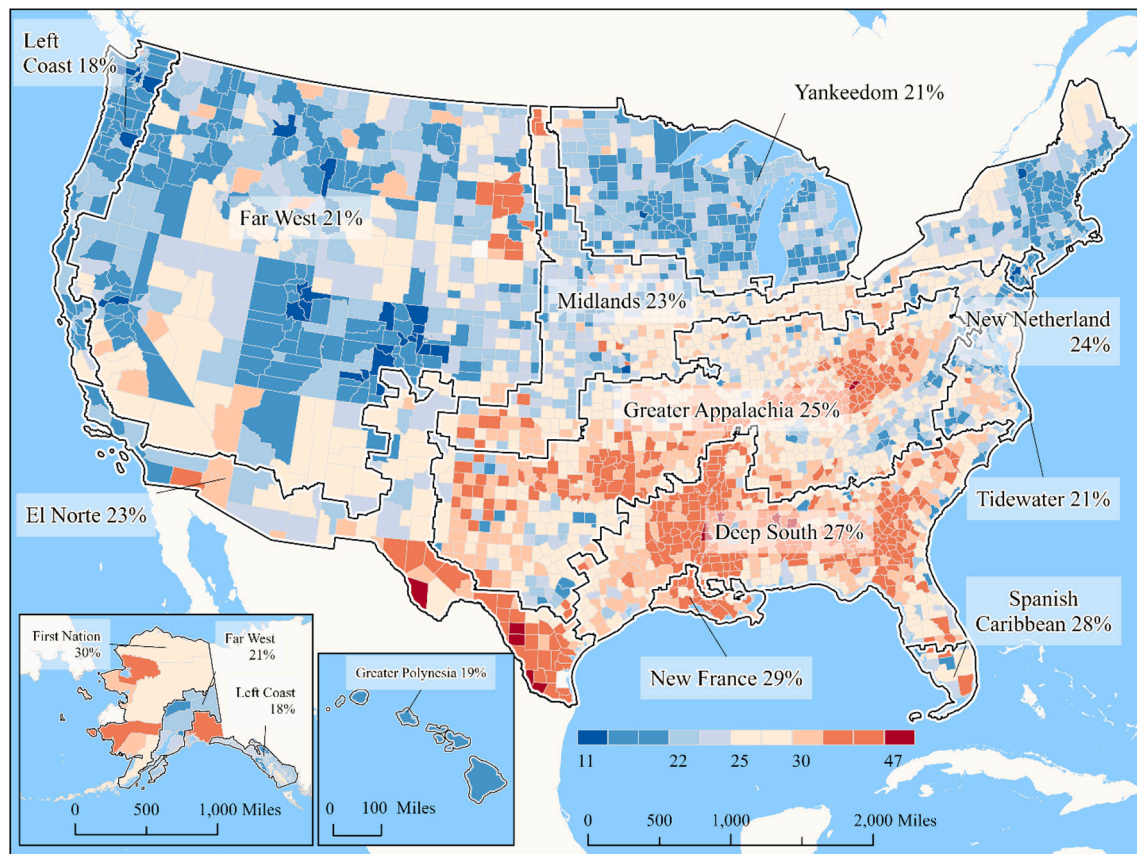
nation is determined by the characteristics of the first group of settlers... regardless of how small the initial band of settlers might have been.” Regional cultures can thus be discerned and mapped by tracking competing first settlement streams, an exercise that has informed the work of historians<sup>20,21</sup> and geographers.<sup>22–24</sup> Woodard<sup>17</sup> applied the concept to posit that North America north of the 25th parallel consists of 11 major regional cultures. This American Nations model has been applied to explain differences in entrepreneurship,<sup>25</sup> economic development,<sup>26</sup> mortality,<sup>27</sup> gender wage gaps,<sup>28</sup> personality characteristics,<sup>29</sup> gun violence,<sup>18</sup> and voting behavior.<sup>30</sup> The unique identities of the *American Nations* are listed in Table 1.

The most recently available U.S. data (2020–2022) on the prevalence of physical inactivity, obesity, and DM, as well as access to exercise opportunities, were obtained from the County Health Rankings & Roadmaps program of the University of Wisconsin Population Health Institute (<https://www.countyhealthrankings.org/>), funded by the Robert Wood Johnson Foundation.<sup>31</sup> The source of data for physical inactivity, obesity, and DM prevalence was the Behavioral Risk Factor Surveillance System (2020: <https://www.cdc.gov/brfss/index.html>)<sup>32</sup> while the source of data for adequate access to exercise opportunities was obtained through a combination of ArcGIS Business Analyst and Living Atlas of the World, YMCA data and U.S. Census TIGER/Line Shapefiles (2022 and 2020: <https://www.countyhealthrankings.org/explore-health-rankings/county-health-rankings-model/health-factors/health-behaviors/diet-and-exercise/access-to-exercise-opportunities?year=2023>).<sup>33</sup> The County Health Rankings defines adequate access to exercise opportunities as “census blocks where the border is a half-mile or less from a park, one mile or less from a recreational facility in an urban area, or 3 miles or less from a recreational facility in a rural area in 2022”. All four measures of interest are reported as percentages in the 2023 County Health Rankings National Database. Using zip-code data, the County Health Rankings database was linked to the *American Nations* dataset from the Nationhood Lab, Pell Center for International Relations and Public Policy, Salve Regina University (<https://www.nationhoodlab.org/>).<sup>18</sup> Percentages for health metrics of interest were then estimated using the population of each *American Nation* listed in Table 1 (Note: Greater Polynesia also included in database linking).

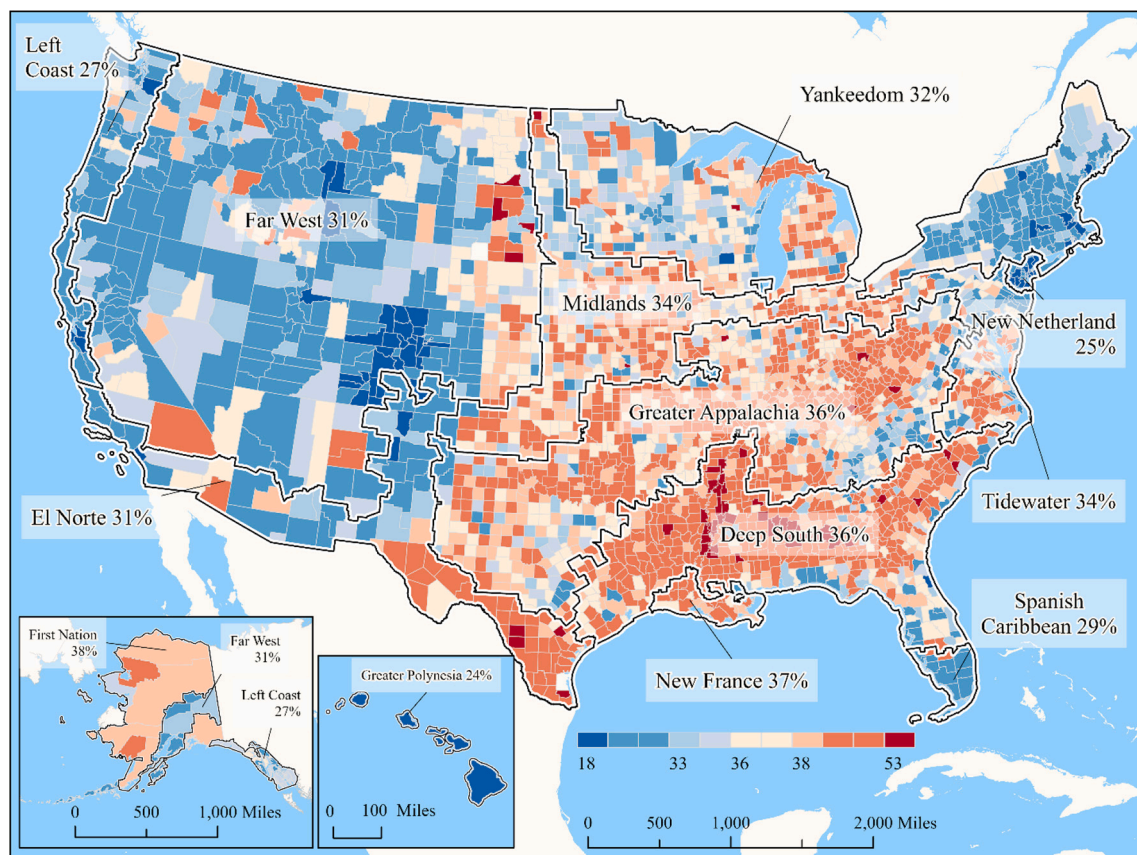
Figs. 1–4 illustrate the differences in physical inactivity, obesity, DM, the three of which commonly cluster together,<sup>34</sup> and adequate access to exercise opportunities among the *American Nations*, with clear trends in prevalence differences becoming readily apparent. Overall, the Deep South and Greater Appalachia have a higher prevalence of physical inactivity, obesity, DM, as well as the lowest adequate access to exercise opportunities.

The maps presented in this commentary provide a novel approach to demonstrating the heterogeneous geography of key U.S. health concerns in the context of a cultural paradigm. Chronic unhealthy lifestyle behaviors, of which physical inactivity is a primary characteristic, initiates a cascade of subsequent events that ultimately leads to a downward spiral in health – obesity and DM are common in those who lead a sedentary life. It is therefore not surprising that the highest prevalence of these unhealthy characteristics cluster together. The fact that the lowest opportunities to exercise likewise exist in these same areas is also not surprising. The maps provided in this commentary indicate there may be an interplay between regional culture and health.

Important questions moving forward relate to *why unhealthy lifestyle characteristics and chronic conditions are heterogeneously distributed across the U.S.*? Certainly, increasing adequate access to exercise opportunities would be an important part of addressing physical inactivity. That is of

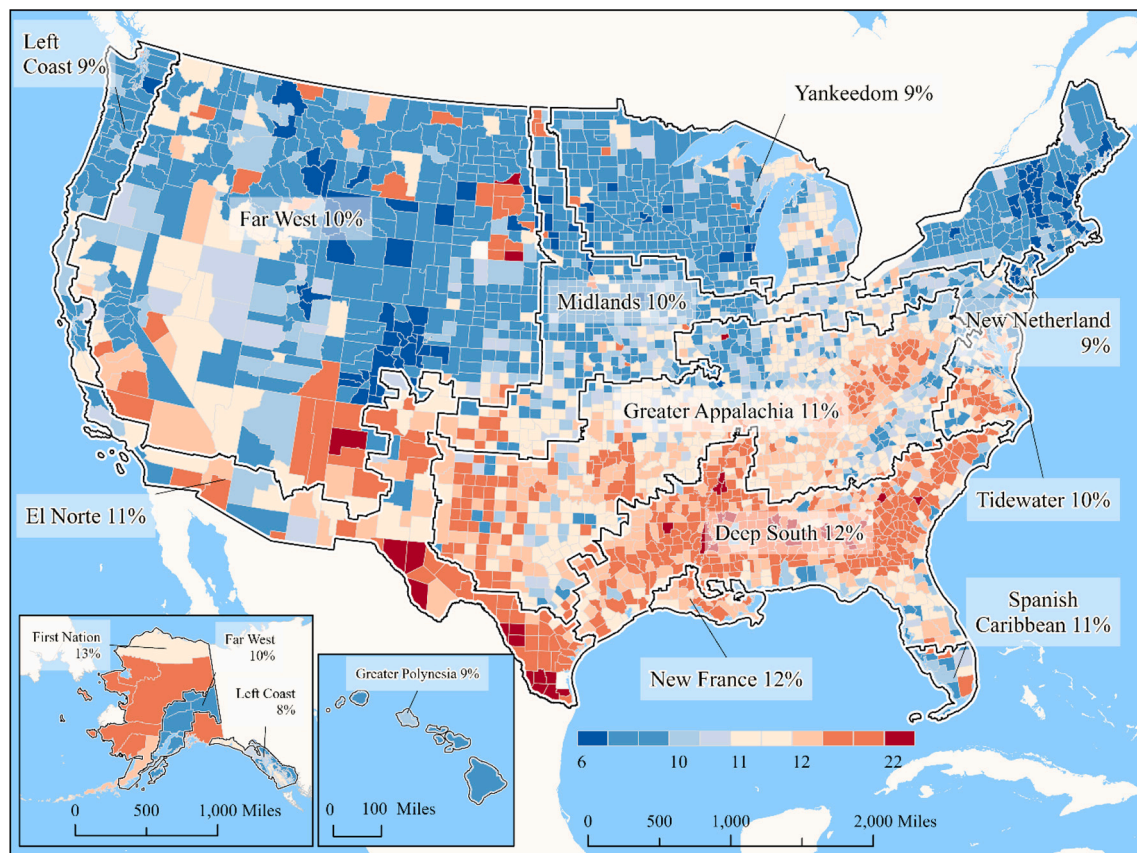


**Fig. 1.** Prevalence of Physical Inactivity.  
John Liberty/Motiv for Nationhood Lab.

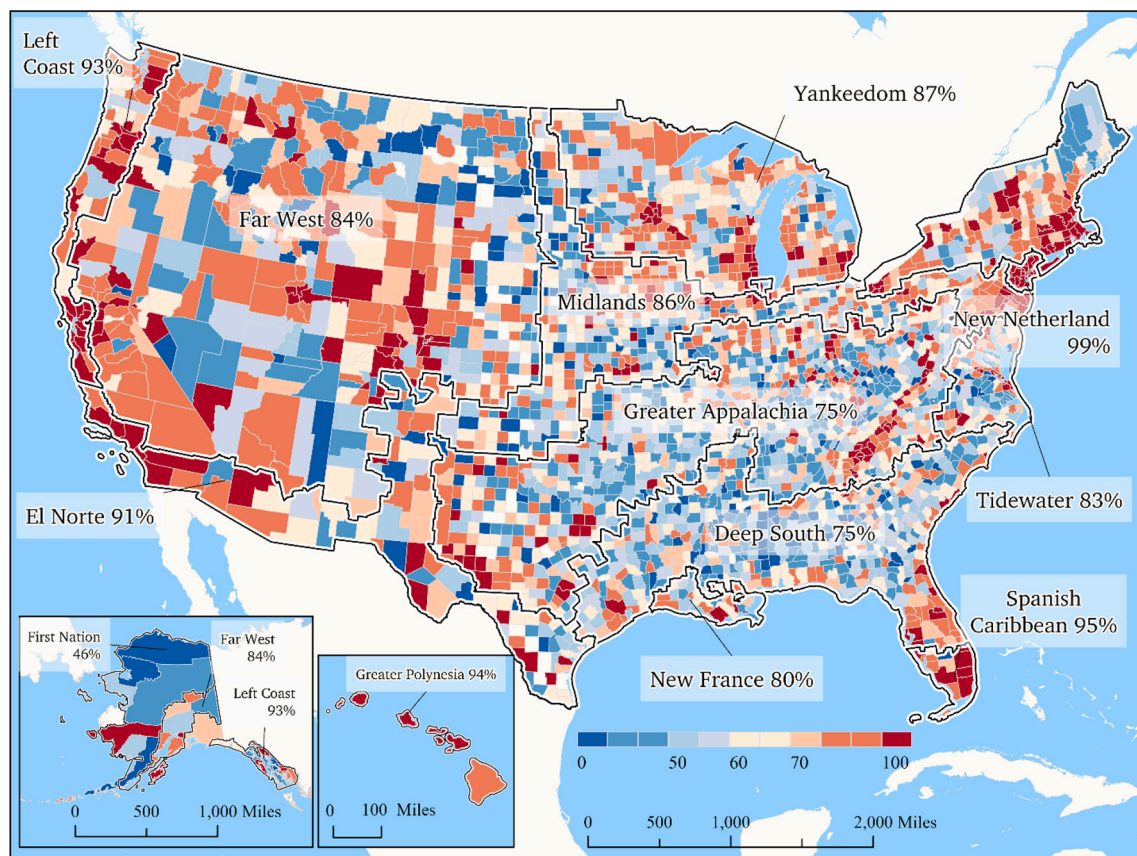


**Fig. 2.** Prevalence of Obesity.  
John Liberty/Motiv for Nationhood Lab.





**Fig. 3.** Prevalence of Diabetes.  
John Liberty/Motiv for Nationhood Lab.



**Fig. 4.** Adequate Access of Exercise Opportunities.  
John Liberty/Motiv for Nationhood Lab.

course only one facet of a multidimensional, complex issue.<sup>35</sup> Perhaps there is something to be learned through viewing the unhealthy living and chronic disease pandemics through the *American Nations* model, specifically helping us understand how stakeholders should uniquely craft the HL message in the context of the regional cultural differences. We are purposely refraining from drawing conclusions from our initial observations – our hope is this approach spurs further discussion and inquiry. Moreover, while there are certainly *hot spots* of particular concern for physical inactivity, obesity, DM, and inadequate access to exercise, there is enormous room for improvement in all *American Nations*. We should strive for optimizing HL behaviors and minimizing the risk of preventable chronic conditions regardless of U.S. geographic region. The key message of the current commentary is the U.S., based on region, has significant differences in cultural history and viewpoints; we must consider these differences when crafting our approach to HL messaging and interventions—whether these are programs, practices, policies, or systems interventions. Moreover, for regions with better HL characteristics and a lower prevalence of chronic disease, we should explore common themes for these findings that transcend cultural differences. As a nation, while there are views, beliefs, and historical experiences that divide us, there are many unifying characteristics that should be identified and promoted. We truly hope readers find our observations thought provoking and spur further exploration to achieve a healthier world for all, especially reductions in CVD risk factors and CVD.

#### Declaration of Competing Interest

None.

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